

# RBE 502 Spring FINAL PROJECT

## Adaptive Control

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**Dynamics of the robotic manipulator** Consider the joint-space dynamics of the OpenManipulator-X robotic manipulator:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} \quad (1)$$

where

$$\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau} \in \mathbb{R}^4,$$

$\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{4 \times 4}$  is the inertia matrix,  $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{4 \times 4}$  is the Coriolis and centrifugal matrix, and  $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^4$  is the gravity vector.

The functions required to evaluate the robot dynamics are provided in the repository through:

- `M_fun.m`
- `C_fun.m`
- `G_fun.m`

The matrices  $\mathbf{M}(\mathbf{q})$ ,  $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ , and  $\mathbf{G}(\mathbf{q})$  are only approximations of the real dynamics of the robot.

The manipulator dynamics can also be written in the linearly parameterized regressor form

$$\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}) \boldsymbol{\pi} = \boldsymbol{\tau}, \quad (2)$$

where  $\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$  is the known regressor matrix, constructed from joint positions, velocities, and accelerations, and  $\boldsymbol{\pi}$  is the constant vector of inertial parameters of the manipulator.

**Project Description** Based on the robotic manipulator model, you must design, implement, and validate an adaptive control law. You must compare the adaptive controller with the controllers developed in previous assignments, such as:

- the computed torque controller,
- the robust controller.

You must clearly present your proposed adaptive controller, including:

- the control law,
- the adaptation law,
- any modifications made to the provided model.

In addition to simulation, you must implement your controllers on the **real robot** and document the tracking errors in both simulation and real-robot experiments. If your controllers cannot be successfully deployed on the real robot, you must clearly explain why and document the reason in your report. **You will be presenting your implementation results during demo day.**

**Required experiments** You must evaluate your controllers in the following two scenarios:

1. **Pose regulation:** regulate the robot to a constant desired joint configuration.
2. **Trajectory tracking:** track a desired joint trajectory  $\mathbf{q}_d(t)$ , together with its corresponding desired velocity  $\dot{\mathbf{q}}_d(t)$  and desired acceleration  $\ddot{\mathbf{q}}_d(t)$ .

You are free to define:

- the desired joint configuration for the regulation task,
- the desired joint trajectories, velocities, and accelerations for the tracking task.

For each of the two tasks above, you must provide:

- simulation results,
- real-robot results.

If the controller does not work on the real robot for one or both tasks, you must still:

- show the simulation results,
- explain the reason for the failure on the real platform
- present these findings during demo day.

**Required results** For each controller and for each experiment, include figures showing:

1. desired joint positions and current joint positions,
2. desired joint velocities and current joint velocities,

3. control torques,
4. the evolution of the tracking error. Quantify the tracking performance using the Euclidean norm of the joint error:

$$\|\mathbf{e}(t)\|_2 = \|\mathbf{q}_d(t) - \mathbf{q}(t)\|_2.$$

Since you are implementing an adaptive control law, you must also include a figure showing the evolution of the estimated parameters.

**Discussion** Your final project report must include a discussion section in which you:

- discuss tracking performance in both regulation and trajectory tracking tasks for each controller,
- compare the behavior observed in simulation and on the real robot,
- explain any gap between the nominal model and the real-robot behavior,
- discuss whether friction compensation was necessary for the real robot and why.

**Note for the real robot** If necessary, you may include an additional friction compensation term when implementing the controller on the real robot. However, this friction term must be used **only** for the real-robot experiments and **not** for the simulation results. **If you use friction compensation, apply it in all controllers (adaptive, computed torque, and robust) and clearly state:**

- the form of the friction model,
- where it is added in the controller

We provide an estimation of the friction model in the following file:

- `ViscousFriction_fun.m`

## Final Project requirements

Your report must be no longer than 3 pages. This is non-negotiable: if you exceed this limit, your report will not be graded. Your report `abc_final_project.pdf` must be written in L<sup>A</sup>T<sub>E</sub>X using the IEEE conference format (double column), where `abc` is the portion of your WPI email address before `@wpi.edu`.